In [9]:
```python
from sklearn.feature_extraction.text import CountVectorizer, TfidfVectorizer
from gensim.models import Word2Vec
import pandas as pd
```

In [10]:
```python
corpus = [
    "Natural Language Processing is amazing",
    "Language models can understand text",
    "Processing text with machine learning is powerful"
]
```

In [11]:
```python
count_vectorizer = CountVectorizer()
bow_counts = count_vectorizer.fit_transform(corpus)
```

In [12]:
```python
bow_df = pd.DataFrame(bow_counts.toarray(), columns=count_vectorizer.get_feature_names
print("\nBag-of-Words (Count Occurrence):\n", bow_df)
```

```
Bag-of-Words (Count Occurrence):
    amazing  can  is  language  learning  machine  models  natural  powerful  \
0        1    0   1         1         0        0       0        1         0
1        0    1   0         1         0        0       1        0         0
2        0    0   1         0         1        1       0        0         1

   processing  text  understand  with
0           1     0           0     0
1           0     1           1     0
2           1     1           0     1
```

In [13]:
```python
normalized_bow_df = bow_df.div(bow_df.sum(axis=1), axis=0)
print("\nBag-of-Words (Normalized Count Occurrence):\n", normalized_bow_df)
```

```
Bag-of-Words (Normalized Count Occurrence):
    amazing  can        is  language  learning   machine  models  natural  \
0      0.2  0.0  0.200000       0.2  0.000000  0.000000     0.0      0.2
1      0.0  0.2  0.000000       0.2  0.000000  0.000000     0.2      0.0
2      0.0  0.0  0.142857       0.0  0.142857  0.142857     0.0      0.0

   powerful  processing      text  understand      with
0  0.000000    0.200000  0.000000         0.0  0.000000
1  0.000000    0.000000  0.200000         0.2  0.000000
2  0.142857    0.142857  0.142857         0.0  0.142857
```

In [14]:
```python
tfidf_vectorizer = TfidfVectorizer()
tfidf_matrix = tfidf_vectorizer.fit_transform(corpus)
tfidf_df = pd.DataFrame(tfidf_matrix.toarray(), columns=tfidf_vectorizer.get_feature_r
print("\nTF-IDF:\n", tfidf_df)
```

```
TF-IDF:
    amazing       can        is  language  learning   machine    models  \
0  0.51742  0.000000  0.393511  0.393511  0.000000  0.000000  0.000000
1  0.00000  0.490479  0.000000  0.373022  0.000000  0.000000  0.490479
2  0.00000  0.000000  0.317570  0.000000  0.417567  0.417567  0.000000

   natural  powerful  processing      text  understand      with
0  0.51742  0.000000    0.393511  0.000000    0.000000  0.000000
1  0.00000  0.000000    0.000000  0.373022    0.490479  0.000000
2  0.00000  0.417567    0.317570  0.317570    0.000000  0.417567
```

In [15]:
```
tokenized_corpus = [sentence.lower().split() for sentence in corpus]
```

In [16]:
```
w2v_model = Word2Vec(sentences=tokenized_corpus, vector_size=50, window=3, min_count=1
```

In [17]:
```
print("\nWord2Vec vector for 'language':\n", w2v_model.wv['language'])
```

```
Word2Vec vector for 'language':
 [ 1.56351421e-02 -1.90203767e-02 -4.11062239e-04  6.93839090e-03
 -1.87794690e-03  1.67635437e-02  1.80215649e-02  1.30730104e-02
 -1.42324448e-03  1.54208085e-02 -1.70686729e-02  6.41421322e-03
 -9.27599426e-03 -1.01779131e-02  7.17923651e-03  1.07406760e-02
  1.55390259e-02 -1.15330126e-02  1.48667190e-02  1.32509898e-02
 -7.41960062e-03 -1.74912829e-02  1.08749345e-02  1.30195096e-02
 -1.57510280e-03 -1.34197138e-02 -1.41718527e-02 -4.99412045e-03
  1.02865072e-02 -7.33047491e-03 -1.87401194e-02  7.65347946e-03
  9.76895820e-03 -1.28571270e-02  2.41711619e-03 -4.14975639e-03
  4.88042824e-05 -1.97670180e-02  5.38400654e-03 -9.50021297e-03
  2.17529293e-03 -3.15245148e-03  4.39334381e-03 -1.57631543e-02
 -5.43437013e-03  5.32639492e-03  1.06933638e-02 -4.78302967e-03
 -1.90201905e-02  9.01175477e-03]
```

In [18]:
```
print("\nMost similar to 'processing':\n", w2v_model.wv.most_similar('processing'))
```

```
Most similar to 'processing':
 [('powerful', 0.16563549637794495), ('amazing', 0.1551763266324997), ('understand', 
0.1393885761499405), ('text', 0.1267007291316986), ('models', 0.12119626253843307), 
('machine', 0.08872979879379272), ('is', 0.011071926914155483), ('learning', -0.02784
1366827487946), ('natural', -0.03339875116944313), ('can', -0.0372748374938 96484)]
```

In [ ]:
```

```